

5.5.

1	2	3	4
5	6	7	
	8	9	
		10	

$$\begin{aligned}
 & \sum_{k=1}^{i-1} (n-k+1) \\
 &= \frac{(n+n-i+2) \cdot (i-1)}{2} \\
 &= \frac{(2n-i+2)(i-1)}{2}
 \end{aligned}$$

$$k = \frac{2ni - i^2 + 2i - 2n + i - 2}{2} + j - i + 1$$

$$= \frac{2ni - i^2 + i}{2} + j - n$$

$$f_1(i) = \frac{2ni - i^2 + i}{2} \quad f_2(j) = j. \quad C = n.$$

5.7.

1	2		
3	4	5	
	6	7	8
		9	10

$$(1). \quad i: 3(i-2) + 2$$

$$j - (i-2)$$

$$\begin{aligned}
 & 3i - 4 + j - i + 2 \\
 & 2i + j - 2
 \end{aligned}$$

$$k = 2i + j - 2$$

$$(2). \quad i = \left\lfloor \frac{k}{3} \right\rfloor + 1$$

$$j = \left\lfloor \frac{k}{3} \right\rfloor + (k \bmod 3)$$

J.8.

0	1			
2	3			
	4	5		
	6	7		
		8	9	
		10	11	
			12	
			13	

$$k = 2(i-1) + 2 - (j \bmod 2) - 1$$

$$= 2i - (j \bmod 2) - 1$$

